Mansha Ul Haq

Aspiring AI/ML Engineer

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SUMMARY

AI/ML Engineer and CS student with hands-on experience in machine learning, deep learning (CNNs, RNNs, LSTMs), Generative and Agentic AI. Proficient in Python, TensorFlow, PyTorch, Scikit-learn, and Transformer architectures; practical exposure to AWS (S3, EC2, SageMaker), GCP (Vertex AI, BigQuery), MLflow, Docker, and CI/CD pipelines. Experienced in MLOps, full-stack RESTful API development, hyperparameter tuning. Strong communicator and collaborative team player with proven analytical problem-solving skills.

WORK EXPERIENCE

Ezitech Jul 2025 – Sep 2025
Machine Learning Intern Rawalpindi, Pakistan

- Optimized data processing workflows, reducing processing time by 15% while deploying robust ML models that achieved 90%+ predictive accuracy.
- Implemented RFE and Randomized Search hyperparameter tuning, resulting in a 20% reduction in training time; meticulously tracked all experiments and model versions using MLflow.
- Designed and optimized deep learning architectures (CNNs, LSTMs) in TensorFlow/Keras; utilized ResNet50 for image data preprocessing, reducing manual preprocessing effort by 30%.
- Conducted comprehensive EDA using statistical techniques and A/B testing; developed and presented data-driven findings to stakeholders, driving a 5% increase in data-driven strategy adoption.

PROJECTS

CDA Field Worker App — Civic Complaint AI System | *Flutter, FastAPI, Python, Multi-Agent Pipeline* [GitHub]

- Built an end-to-end Flutter + FastAPI system enabling municipal field workers to ingest, dispatch, and resolve citizen complaints in real time, unifying scraped social-media reports and manual entries through a single ingestion pipeline.
- Engineered a 5-stage AI reasoning pipeline (observation, reasoning, constraint-checking, adaptation, action) generating transparent, auditable decision traces rendered as collapsible UI components for field accountability.
- Designed a singleton state-management architecture (`ChangeNotifier`) with a real-time status timeline and SLA tracking, ensuring overflow-safe, WCAG AA-compliant UI across all device sizes.
- Delivered an extensible REST API (FastAPI) supporting full work-order lifecycle management (dispatch, status updates, resolution reporting), built as a drop-in mock layer swappable for a production backend without frontend changes.

Real-Time Helmet Detection System | *Python, YOLOv8, OpenCV, Roboflow, Streamlit, Docker* [GitHub]

- Developed a real-time helmet detection system using YOLOv8 with image augmentation (e.g., resizing to 640, grayscale, brightness), achieving 77.1% mAP@0.5, 86% compliance accuracy, and a 95% detection rate; evaluated using F1-score, Precision, Recall, and AUC-ROC metrics.
- Engineered and deployed the system via a Dockerized Streamlit application for real-time inference, ensuring continuous safety compliance monitoring in industrial environments.

Social Media Sentiment Analysis Pipeline | *Python, BERT, Transformers, Hugging Face, FastAPI, SHAP* [GitHub]

- Built a social media sentiment analysis model achieving 92% classification accuracy; deployed as a FastAPI RESTful microservice endpoint for real-time inference.
- Processed and analyzed social media data to generate dashboards surfacing actionable consumer insights for marketing strategies, utilizing SHAP and LIME to improve model interpretability by 20%.

EduBot — AI Academic Assistant | *Python, Gemini API, RAG, LangChain, Streamlit, Git* [GitHub]

- Designed and implemented an AI academic assistant, improving response relevance by 40% and integrating advanced prompt engineering to cut hallucination rates by 15%.
- Facilitated seamless deployment via Streamlit with Git/GitHub Actions CI/CD for continuous integration and delivery, supporting multimodal capabilities for richer query context.

EDUCATION

University of Engineering & Technology Sep 2023 – Jun 2027
B.S. Computer Science Mardan, Pakistan
Coursework: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Machine Learning, Deep Learning, Calculus, Linear Algebra, Probability & Statistics, DBMS, Artificial Intelligence

SKILLS

Programming Languages: Python, C/C++, Dart, JavaScript
Frameworks & Libraries: TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, OpenCV, LangChain, Transformers, YOLO, BERT, Flutter, Django
Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebooks
Databases: MySQL, PostgreSQL, MongoDB, Firebase